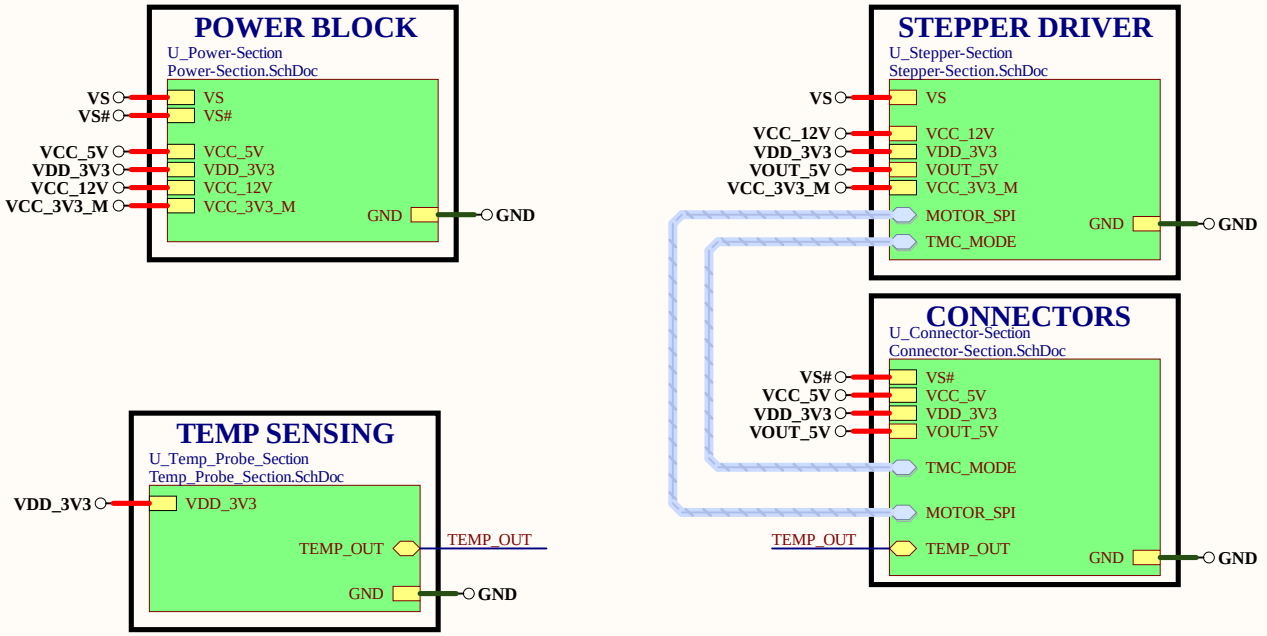
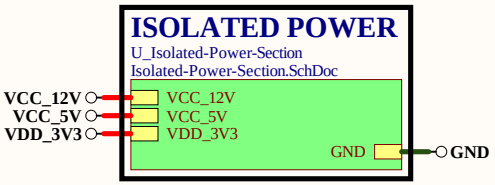
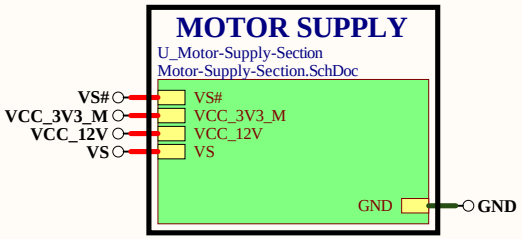
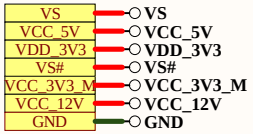
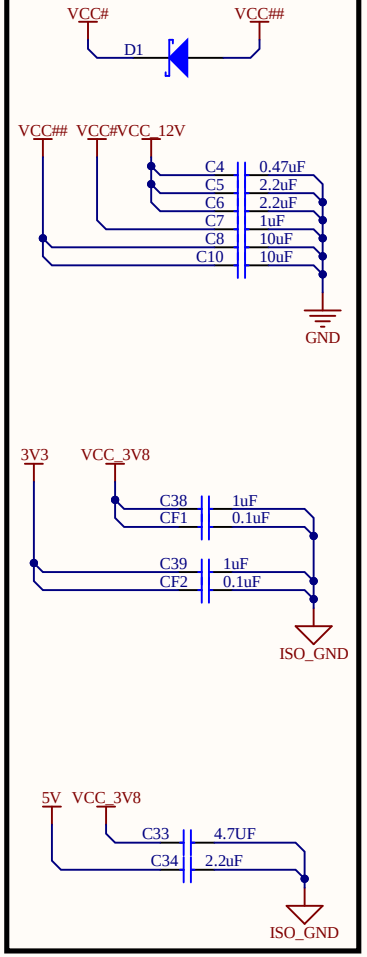

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VS#
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS#
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND
GND

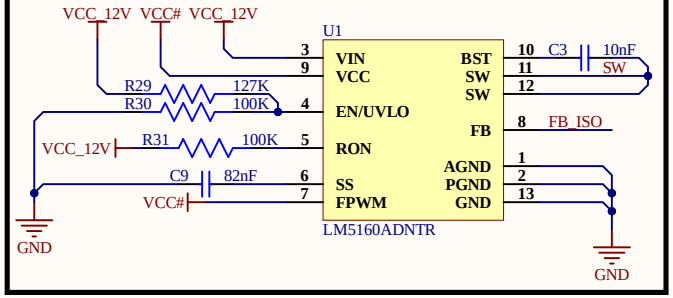

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VS#
VCC_3V3_M
VCC_12V
VS
VS#
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

DECOUPLING CAPS

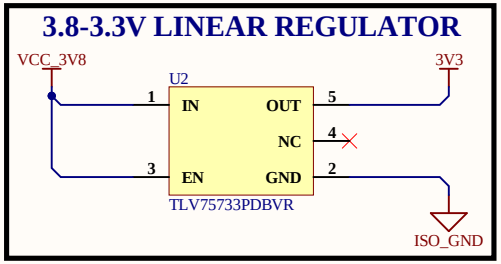


12V-3.8V ISOLATED BUCK CONVERTER

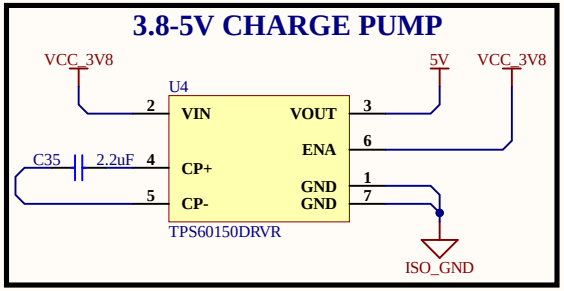


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

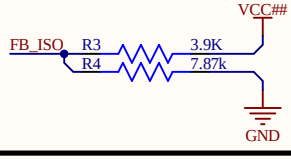
3.8-3.3V LINEAR REGULATOR



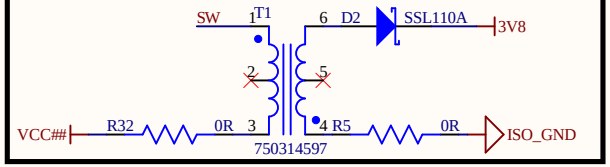
3.8-5V CHARGE PUMP



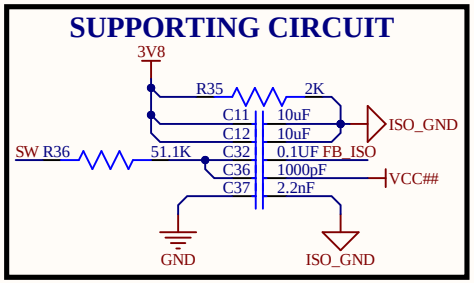
FEEDBACK NETWORK



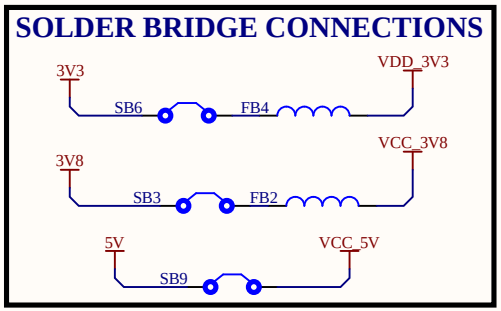
FLYBACK TRANSFORMER

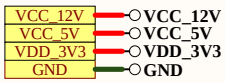


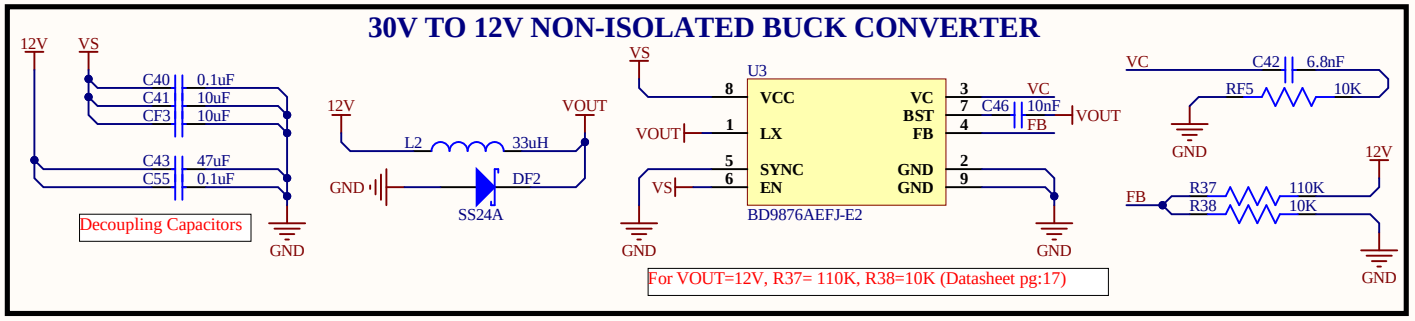
SUPPORTING CIRCUIT

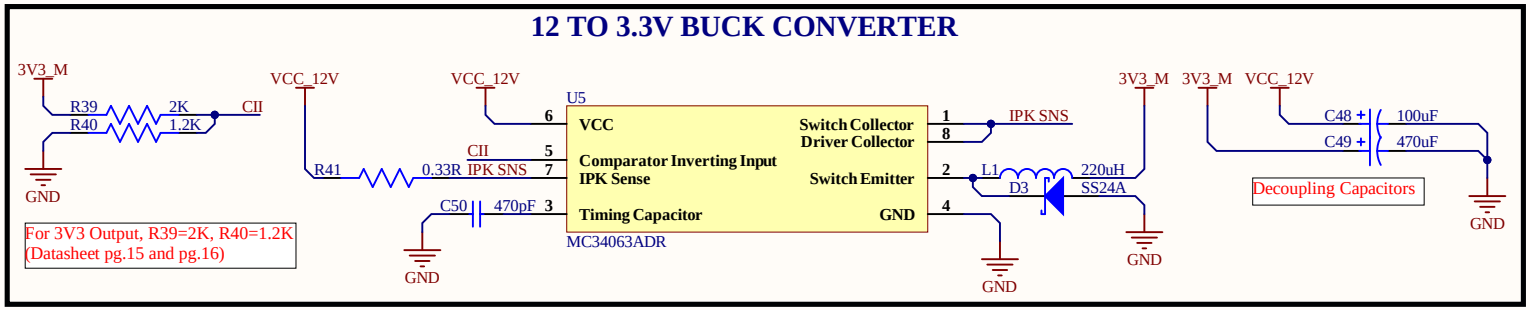


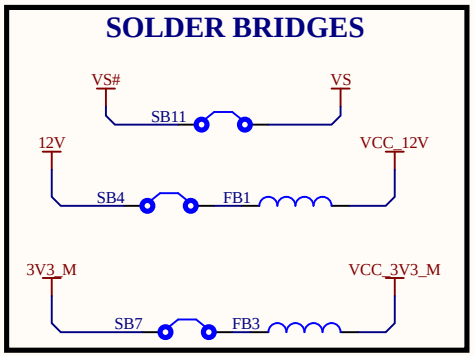
SOLDER BRIDGE CONNECTIONS

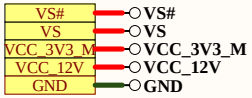


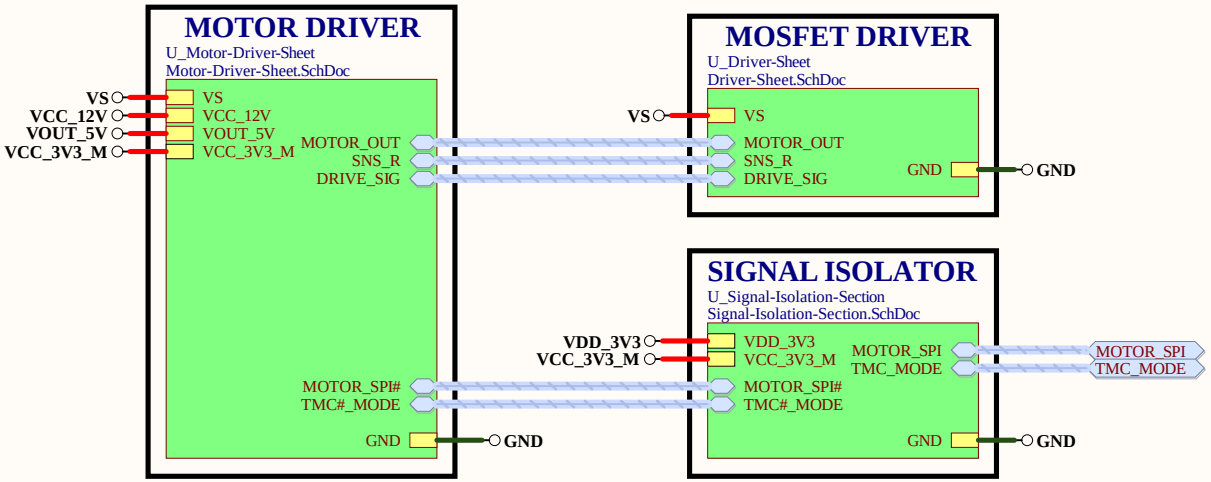


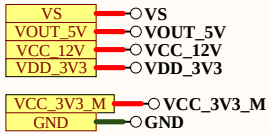

30V TO 12V NON-ISOLATED BUCK CONVERTER
12V
VS
C40 0.1uF
C41 10uF
CF3 10uF
C43 47uF
C55 0.1uF
Decoupling Capacitors
GND
12V
VOUT
L2 33uH
GND
DF2
SS24A
VS
U3
VCC
LX
SYNC
EN
VC
BST
FB
GND
GND
BD9876AEFJ-E2
VOUT
C46 10nF
VC
C42 6.8nF
RF5 10K
GND
FB
R37 110K
R38 10K
12V
GND
For VOUT=12V, R37= 110K, R38=10K (Datasheet pg:17)

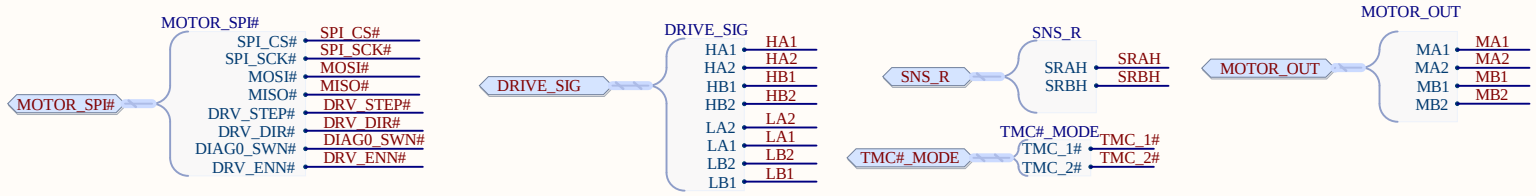

12 TO 3.3V BUCK CONVERTER
3V3_M
R39 2K
R40 1.2K
CII
GND
For 3V3 Output, R39=2K, R40=1.2K
(Datasheet pg.15 and pg.16)
VCC_12V
R41 0.33R
IPK SNS
CII
C50 470pF
GND
VCC_12V
U5
VCC
Comparator Inverting Input
IPK Sense
Timing Capacitor
Switch Collector
Driver Collector
Switch Emitter
GND
MC34063ADR
IPK SNS
L1 220uH
D3 SS24A
3V3_M
GND
GND
3V3_M
3V3_M
VCC_12V
C48 100uF
C49 470uF
Decoupling Capacitors
GND


SOLDER BRIDGES
VS#
VS
SB11
12V
VCC_12V
SB4
FB1
3V3_M
VCC_3V3_M
SB7
FB3

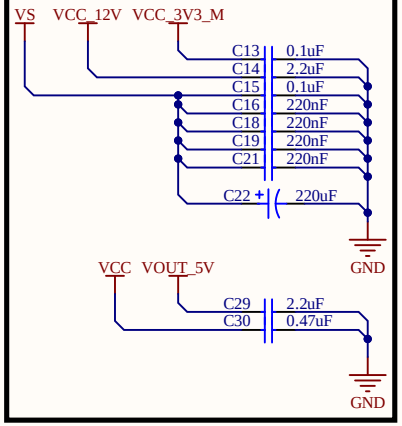

VS#
VS
VCC_3V3_M
VCC_12V
GND
VS#
VS
VCC_3V3_M
VCC_12V
GND


MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
VS
VS
MOTOR_OUT
SNS_R
DRIVE_SIG
GND
GND
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
VCC_3V3_M
VDD_3V3
VCC_3V3_M
MOTOR_SPI
TMC_MODE
MOTOR_SPI#
TMC#_MODE
GND
GND
MOTOR_SPI
TMC_MODE

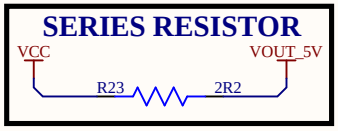

VS
VOUT_5V
VCC_12V
VDD_3V3
VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND
VCC_3V3_M
GND



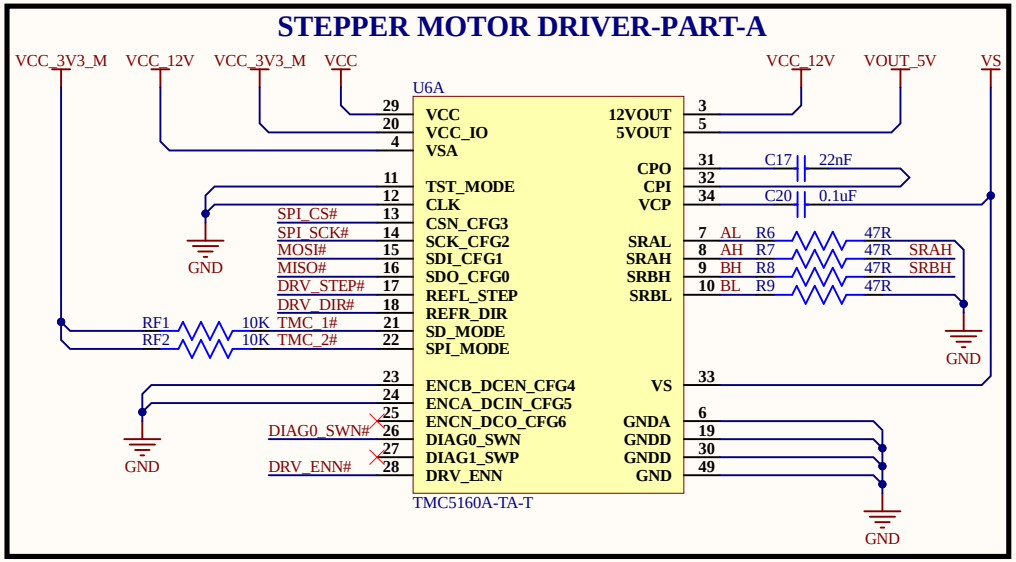
DECOUPLING CAPS



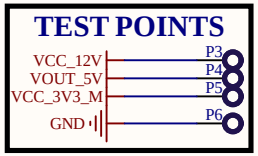
SERIES RESISTOR



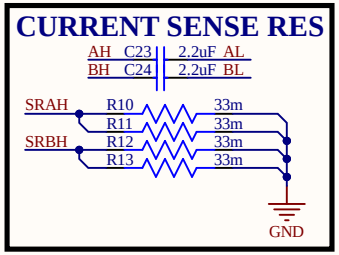
STEPPER MOTOR DRIVER-PART-A



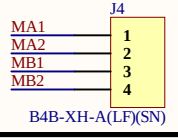
TEST POINTS



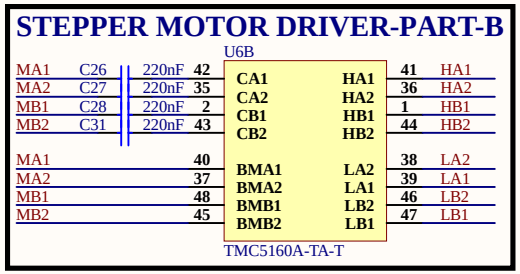
CURRENT SENSE RES



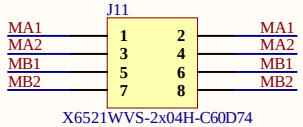
MOTOR OUTPUT

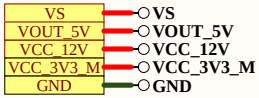


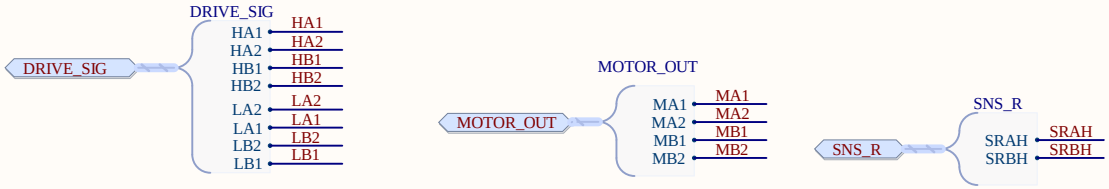
STEPPER MOTOR DRIVER-PART-B



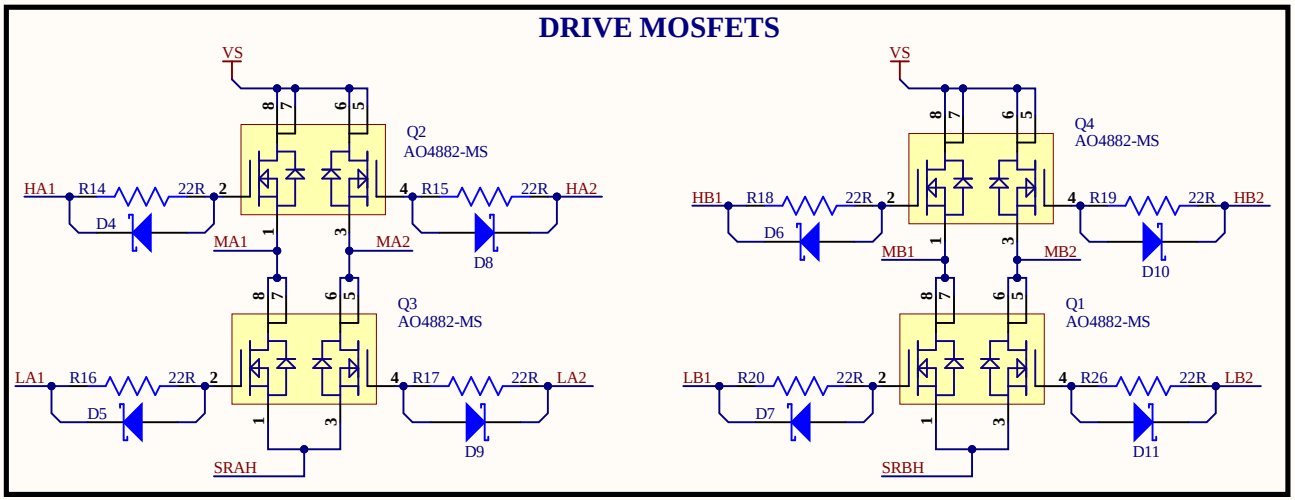
MOTOR OUTPUT FOR NEMA34



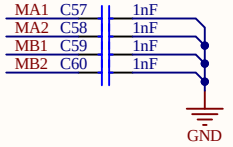




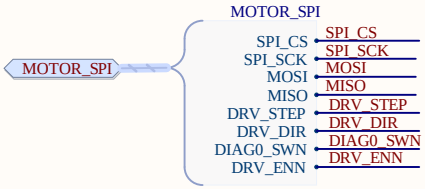
DRIVE MOSFETS

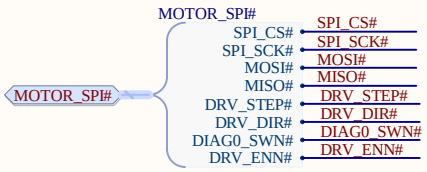


DECOUPLING CAPS



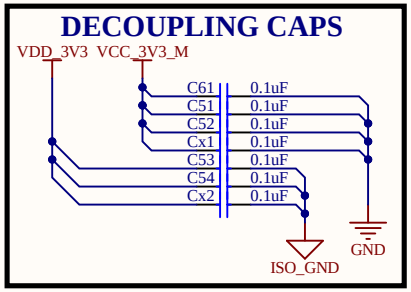


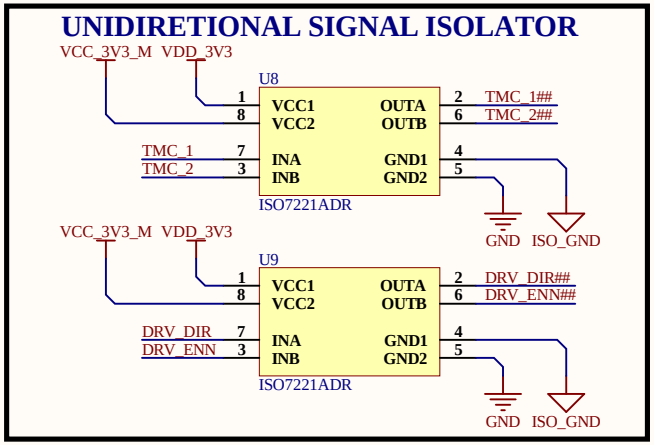


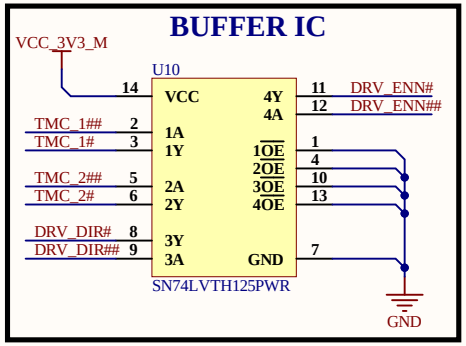


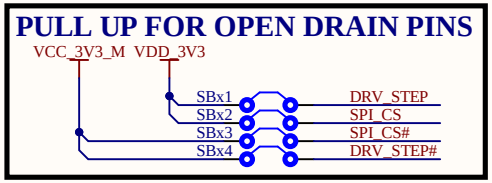




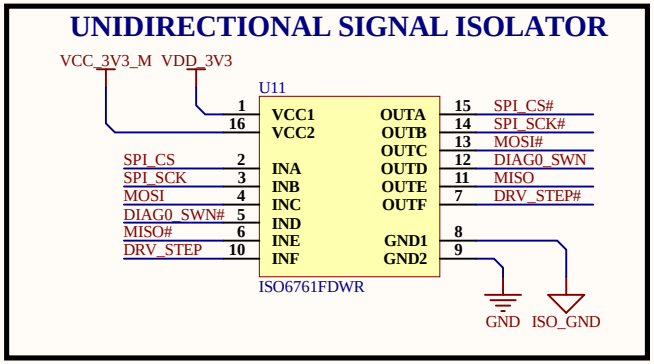








Mount 1K 0603 resistors on these bridges if needed

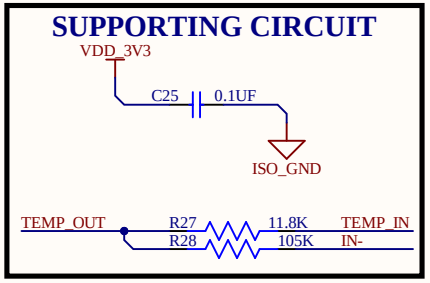




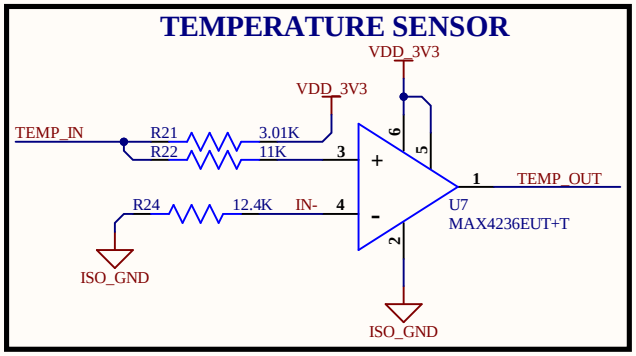
Title		
Size A4	Number	Revision
Date: 2/15/2023		Sheet of
File: E:\ColeBenson-SmartMotors\..\Signal-Isolation-Section.SchDoc		Drawn By:

TEMP_OUT TEMP_OUT

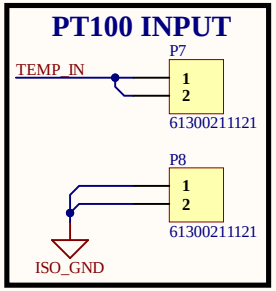
SUPPORTING CIRCUIT



TEMPERATURE SENSOR



PT100 INPUT



VDD_3V3 **VDD_3V3**
GND **GND**

INDICATION FOR MOTOR 5V SUPPLY

TMC_MODE
TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2

TEMP_OUT
TEMP_OUT

VOUT_5V
R25
143R
LED2
GND

3.3V RGB LED

VDD_3V3
LED1
R33 51R 6 1 LED_B
R34 150R 5 2 LED_R
R46 51R 4 3 LED_G
LTST-G683GEBW

13 MM ACTUATOR

SW1
BUTTON
3
4
C2
0.1uF
1
2
Tactile Switch 13mm Actuator
ISO_GND

HEADERS TO SELECT BOOT MODE

VDD_3V3
RF3 10K BOOT1
RF4 10K BOOT0
Default: Embedded SRAM Mode

P9
1
2
BOOT1
61300211121
ISO_GND

P10
1
2
BOOT0
61300211121
ISO_GND

GND BOOT1 to enter
System Memory Bootloader Mode

GND BOOT0 to enter
User Flash Memory Mode

MOTOR_SPI
MOTOR_SPI
SPI_CS SPI_CS
SPI_SCK SPI_SCK
MOSI MOSI
MISO MISO
DRV_STEP DRV_STEP
DRV_DIR DRV_DIR
DIAG0_SWN DIAG0_SWN
DRV_ENN DRV_ENN

USB TYPE C FOR DATA TRANSFER

J1
A4 A9 B4 B9 VBUS
A8 B8 RFU1
A2 B11 SSTXp1
A3 B10 SSRXn1
USB_DP A6 Dp1
USB_DM B7 Dn1
USB_DM A7 Dn1
USB_DP B6 Dp1
CC1 A5
CC1 B5
SSRXn2 A10
SSTXn2 B3
SSRXn2 A11
SSTXp2 B2
GND A1
GND B1
GND A12
GND B12
MH1
MH2
MP1
MP2
MP3
MP4
R1 5.1K
R2 5.1K
ISO_GND
MC-318FB
ISO_GND

BOARD TO BOARD CONNECTOR

J3
VCC_5V
LED_R A1
USB_DM A2
USB_DP A3
CAN_NL A4
CAN_NH A5
BOOT0 A6
LED_B A7
BUTTON A8
TMC_2 A9
TMC_1 A10
BOOT1 A11
A12
A13
MP1
MP3
MH1
B1 LED_G
B2 SPI_CS
B3 SPI_SCK
B4 MOSI
B5 MISO
B6 DRV_STEP
B7 DRV_DIR
B8 DIAG0_SWN
B9 DRV_ENN
B10 TEMP_OUT
B11
B12
B13
MP2
MP4
MH2
VDD_3V3
ISO_GND
M55-7012642R
ISO_GND

DECOUPLING CAP

VS#
C1 220uF
GND

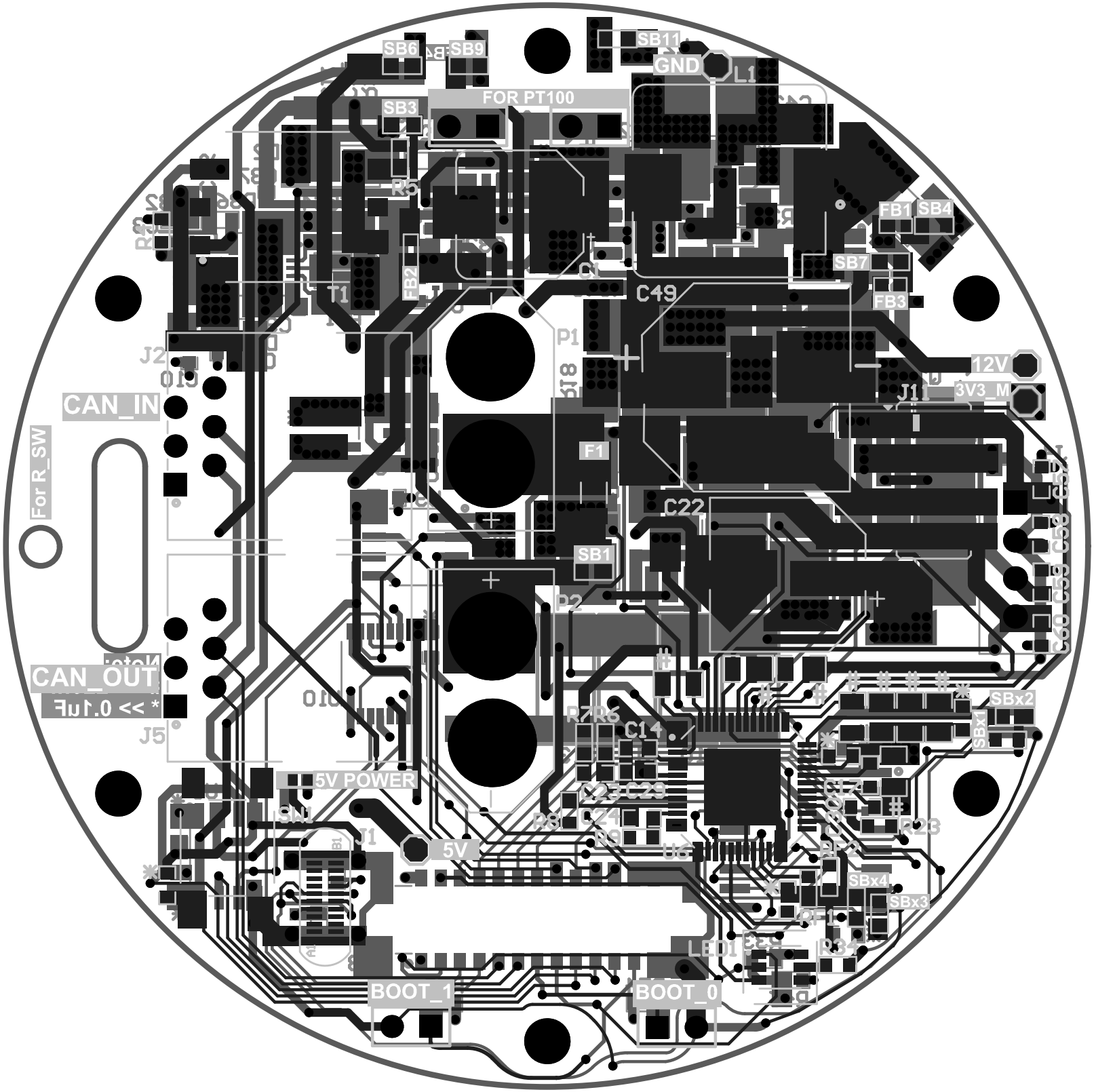
INPUT CONNECTORS

VS#
F1
P1
2
1
PRT-10474
GND
VS#
SB1
P2
2
1
PRT-10474
GND

RJ11 FOR CAN BUS

J2
CAN_NL 1
2
CAN_NH 3
4
5
6
MH1
MH2
5520425-3
CAN IN
J5
CAN_NH 1
2
CAN_NL 3
4
5
6
MH1
MH2
5520425-3
CAN OUT

VS# VS#
VCC_5V VCC_5V
VDD_3V3 VDD_3V3
VOUT_5V VOUT_5V
GND GND


SB6
SB9
SB11
GND
L1
FOR PT100
SB3
R5
FB1
SB4
SB7
FB3
FB2
T1
C49
P1
J2
12V
3V3_M
J11
CAN_IN
For R_SW
F1
C22
SB1
P2
CAN_OUT
* >> 0.1uF
SBx2
SBx1
R7R6
C14
5V POWER
C23C29
R8
C24
R9
J1
5V
SBx4
SBx3
LED1
R34
BOOT_1
BOOT_0